Math 661 Applied Statistics

Module 2 Assignment Solutions

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Task 1: Gosset & Guinness — IPS Exercises 1.37, 1.38, 1.39

William Sealy Gosset worked at the Guinness Brewery in Dublin and made substantial contributions to the practice of statistics. In one study, Gosset examined the change in the double stout market before and after World War I (1914–1918). For various regions in England and Scotland, he calculated the ratio of sales in 1925 (after the war) as a percent of sales in 1913 (before the war).

Data: Bristol 94, Cardiff 112, English Agents 78, English O 68, English P 46, English R 111, Glasgow 66, Liverpool 140, London 428, Manchester 190, Newcastle-on-Tyne 118, Scottish 24

Exercise 1.37: Measures of Center

(a) Mean:

$x\blacksquare = (1/n) \times \Sigma x_i = (1/12) \times (94 + 112 + 78 + 68 + 46 + 111 + 66 + 140 + 428 + 190 + 118 + 24) = (1/12) \times 1475 =$ **122.92**

(b) Median:

Sorted data: 24, 46, 66, 68, 78, 94, 111, 112, 118, 140, 190, 428

With n = 12 (even), the median is the average of the 6th and 7th values: (94 + 111) / 2 = **102.5**

(c) Preferred measure of center: The median (102.5) is preferred over the mean (122.92) because the distribution is strongly right-skewed due to the extreme London value of 428. The mean is pulled upward by this outlier and does not represent the typical sales ratio. The median is resistant to outliers and better represents the center of this distribution.

Exercise 1.38: Measures of Spread

(a) Standard Deviation: s = 105.73

Computed using the sample standard deviation formula: $s = \sqrt{[(1/(n-1)) \times \Sigma(x_i - x\blacksquare)^2]}$

(b) Quartiles:

Q_1 = 67.5 (25th percentile), Q_3 = 123.5 (75th percentile)

Five-Number Summary: Min = 24, Q_1 = 67.5, Median = 102.5, Q_3 = 123.5, Max = 428

(c) Preferred measure of spread: The IQR (56.0) is preferred over the standard deviation (105.73) because the data contains an extreme outlier (London = 428). The standard deviation, like the mean, is sensitive to outliers. The IQR, measuring the spread of the middle 50% of the data, is resistant to outliers and gives a more representative measure of the typical variability.

Exercise 1.39: Outlier Detection and Boxplots

(a) IQR: IQR = $Q_3 - Q_1$ = 123.5 − 67.5 = **56.0**

(b) Outlier Detection (1.5 × IQR Rule):

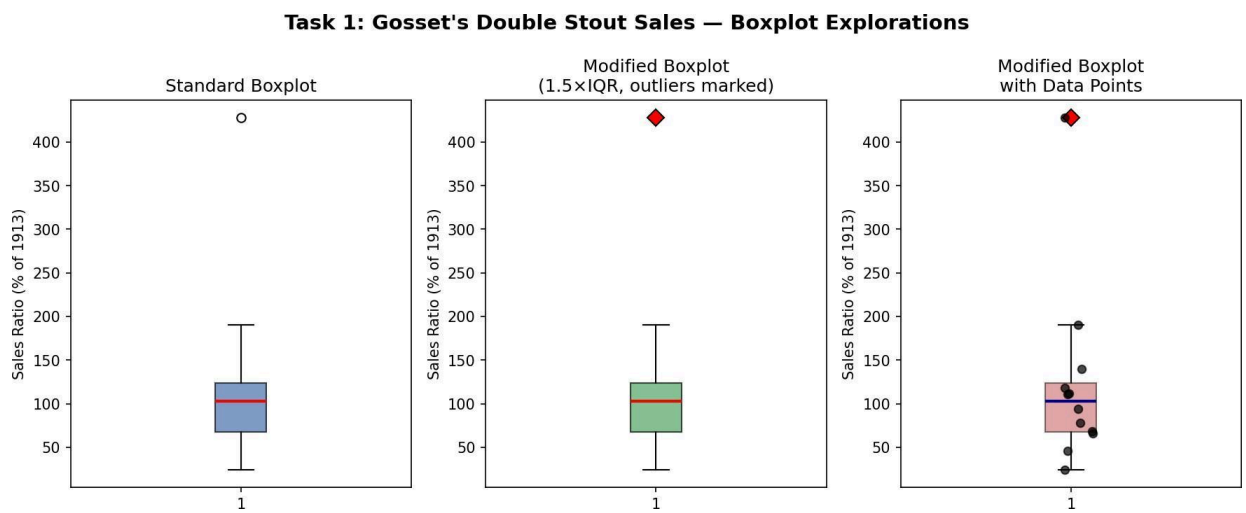
Lower fence = Q_1 − 1.5 × IQR = 67.5 − 1.5 × 56.0 = -16.5

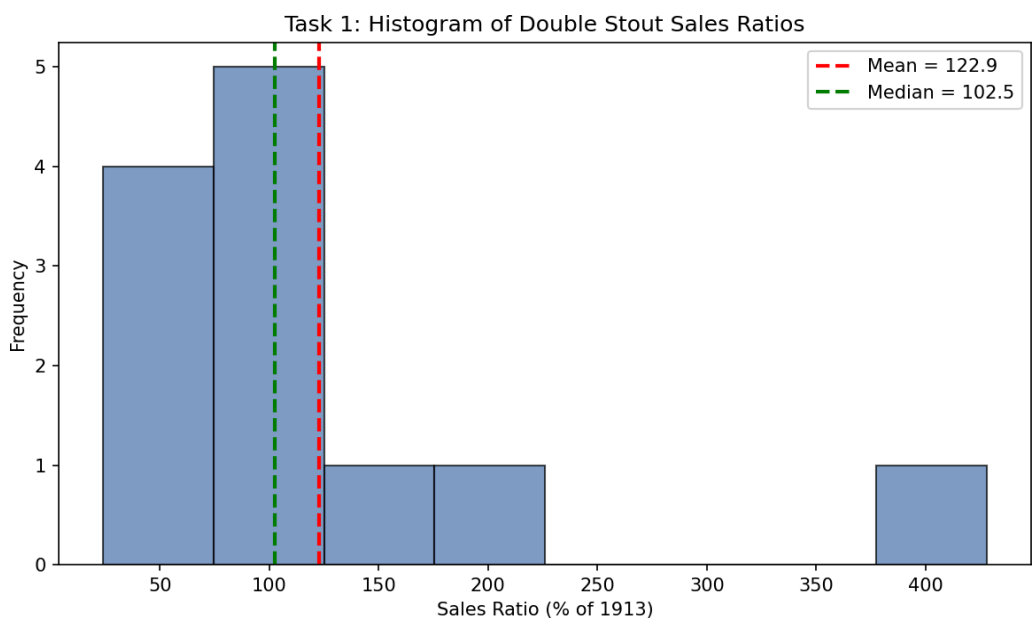
Upper fence = Q_3 + 1.5 × IQR = 123.5 + 1.5 × 56.0 = 207.5

Outlier(s): London (428). These values fall outside the fences [-16.5, 207.5].

(c) –(f) Boxplot Explorations:

Below are three boxplot variations for the double stout data. The standard boxplot shows the basic five-number summary. The modified boxplot explicitly marks outliers beyond 1.5×IQR. The third overlay individual data points to show the actual data distribution.





Task 1: Stemplot of Double Stout Sales

```
Stem | Leaves
-----|----------
   2 | 4
   4 | 6
   6 | 6 8
   7 | 8
   9 | 4
  11 | 1 2 8
  14 | 0
  19 | 0
  42 | 8

Stem unit = 10
Leaf unit = 1
```

Comparison of Graphical Summaries: The standard boxplot is compact and shows the five-number summary clearly but does not distinguish outliers. The modified boxplot explicitly flags outliers (London at 428) and is more informative for identifying unusual observations. The stemplot shows the actual data values and the shape of the distribution more clearly, revealing the strong right-skew caused by the London outlier. The histogram confirms the right-skewed shape. For this dataset, the modified boxplot with overlaid data points provides the best balance of summary information and data transparency.

What I learned: The double stout sales data reveals that the London market had a dramatically different post-war experience compared to all other regions. While most regions saw sales decline or remain roughly stable after WWI (median ratio around 101), London's sales ratio of 428% was an extreme outlier. This single value strongly influences the mean, making the median a better measure of the typical post-war sales change. The wide gap between mean and median is a clear signal of a skewed distribution with outliers.

Task 2: Potassium from Potatoes

Data: Potassium absorption (mg) from 27 adults on a controlled diet including 40 mEq of potassium from potatoes for 5 days.

(a) Stemplot:

Task 2: Stemplot of Potassium Absorption Data

```
Stemplot of Potassium Absorption (mg)
Stem unit = 1000 mg, Leaf unit = 100 mg

 Stem | Leaves
------|-----------------------------------
   26 | 6
   27 | 8
   29 | 5 5 7 7
   30 | 2 4 6 7 9
   31 | 1 1 3 4
   32 | 2 3 5 6 7 8
   34 | 8
   35 | 0 3 8
   36 | 0
   42 | 1

Key: 26|6 represents 2660 mg
```

(b) Distribution Pattern:

The distribution of potassium absorption shows a roughly unimodal pattern with the bulk of observations concentrated between approximately 2950 mg and 3300 mg. The distribution appears slightly right-skewed, with a longer tail extending toward higher values (3500–4200 mg). Most of the data clusters around 3000–3200 mg, suggesting that this is the typical range of potassium absorption from the potato-based diet.

(b) Outlier Analysis:

IQR: $Q_3 - Q_1 = 3283.06 - 3036.62 = 246.44$

Lower fence: $Q_1 - 1.5 \times IQR = 3036.62 - 1.5 \times 246.44 = 2666.95$

Upper fence: $Q_3 + 1.5 \times IQR = 3283.06 + 1.5 \times 246.44 = 3652.73$

Outlier(s) identified: 2664.38 mg (below lower fence); 4213.49 mg (above upper fence). These values fall outside the fence boundaries [2666.95, 3652.73]. The high value of 4213.49 mg represents an unusually high potassium absorption, while 2664.38 mg is marginally below the lower fence, indicating

relatively low absorption for this individual.

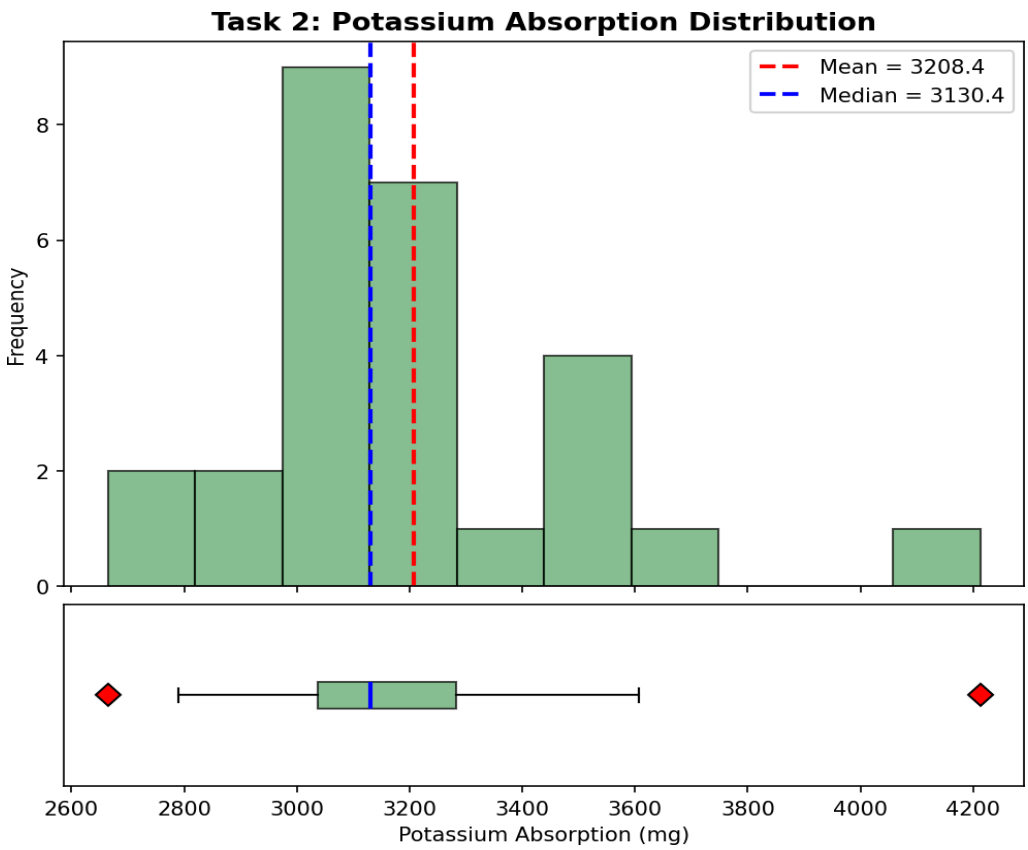
(c) Shape, Center, and Spread:

Shape: The distribution is slightly right-skewed and approximately unimodal. The right-skew is evidenced by the mean (3208.44) being greater than the median (3130.37) and the presence of a few higher values pulling the tail to the right.

Center: The mean potassium absorption is 3208.44 mg and the median is 3130.37 mg. The relatively small difference between mean and median suggests the skew is moderate.

Spread: The standard deviation is 306.68 mg. The IQR is 246.44 mg. The range is 1549.11 mg (from 2664.38 to 4213.49).

Five-Number Summary: Min = 2664.38, Q_1 = 3036.62, Median = 3130.37, Q_3 = 3283.06, Max = 4213.49



Task 3: Trimmed Mean — IPS Exercise 1.59

A trimmed mean is computed by removing a fixed percentage of observations from each end of the sorted data before computing the mean. This provides a measure of center that is resistant to outliers, offering a compromise between the mean and the median.

Double Stout Data (Task 1):

Original Mean: 122.92 | Median: 102.5

5% Trimmed Mean: 122.92 | 10% Trimmed Mean: 102.30

The trimmed means (122.92 and 102.30) are substantially lower than the full mean (122.92), reflecting the influence of the extreme London value. The 10% trimmed mean is closer to the median, confirming that the outlier inflates the regular mean.

Potassium Data (Task 2):

Original Mean: 3208.44 | Median: 3130.37

5% Trimmed Mean: 3190.00 | 10% Trimmed Mean: 3189.30

For the potassium data, the trimmed means are close to both the full mean and the median, indicating that this distribution has fewer influential extreme values than the stout data.

Task 4: Proof — Deviations from the Mean Sum to Zero

Theorem: For any set of observations $x_1, x_2, ..., x_n$ with mean $x\blacksquare$, the sum of deviations from the mean equals zero:

$$\Sigma(x_i - x\blacksquare) = 0$$

Proof:

By definition, the sample mean is: $x\blacksquare = (1/n) \times \Sigma x_i$

Therefore: $n \times x\blacksquare = \Sigma x_i$

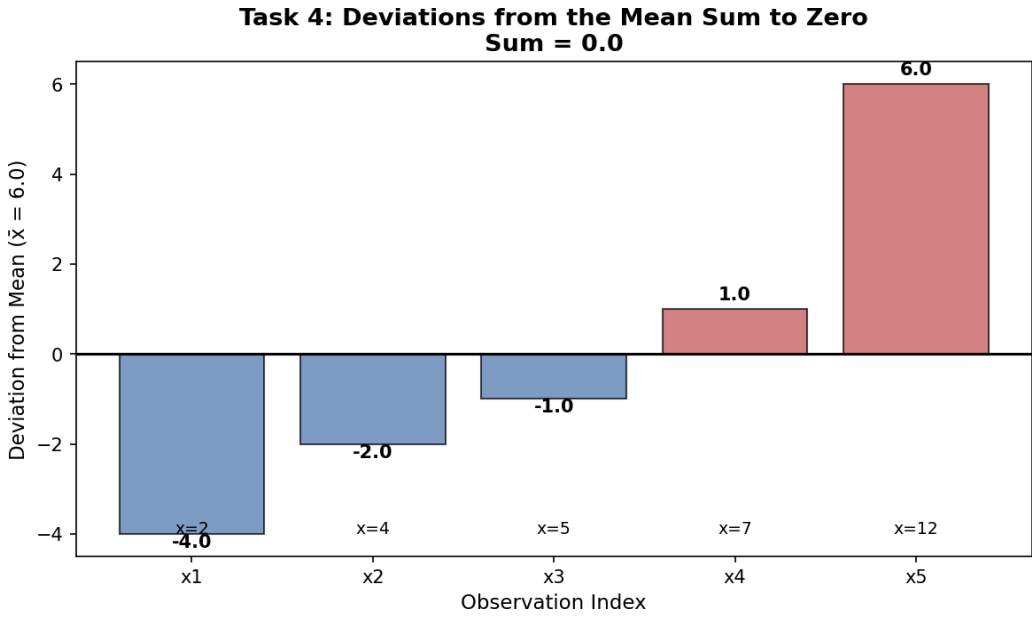
Now consider the sum of deviations:

$$\Sigma(x_i - x\blacksquare) = \Sigma x_i - \Sigma x\blacksquare = \Sigma x_i - n \times x\blacksquare$$

Substituting $n \times x\blacksquare = \Sigma x_i$:

$$\Sigma(x_i - x\blacksquare) = \Sigma x_i - \Sigma x_i = \mathbf{0} \quad \blacksquare$$

This result is one reason why the variance and standard deviation use *squared* deviations: since the raw deviations always sum to zero, they cannot serve as a useful measure of spread. Squaring the deviations makes all terms positive, giving a meaningful measure of variability.

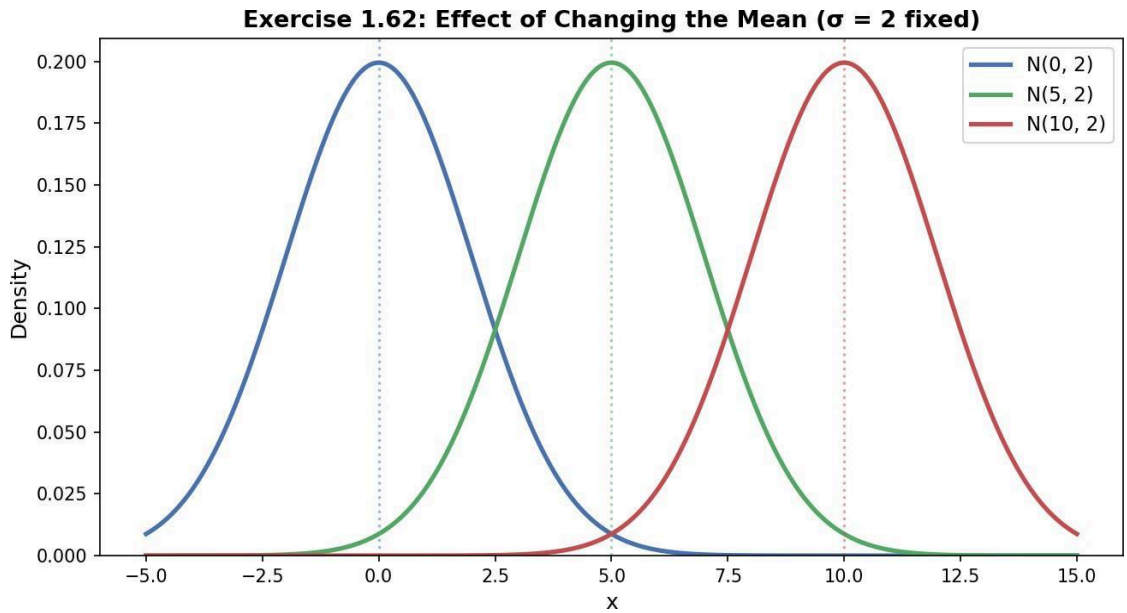

Task 4: Deviations from the Mean Sum to Zero
Sum = 0.0
Deviation from Mean (x̄ = 6.0)
6
4
2
0
−2
−4
6.0
1.0
-1.0
-2.0
-4.0
x=2
x=4
x=5
x=7
x=12
x1
x2
x3
x4
x5
Observation Index

Task 5: Sketching Density Curves — IPS Exercises 1.62, 1.63, 1.64

Normal density curves are completely determined by two parameters: the mean μ (which controls the location/center) and the standard deviation σ (which controls the spread). Below we explore how changing each parameter affects the shape of the curve.

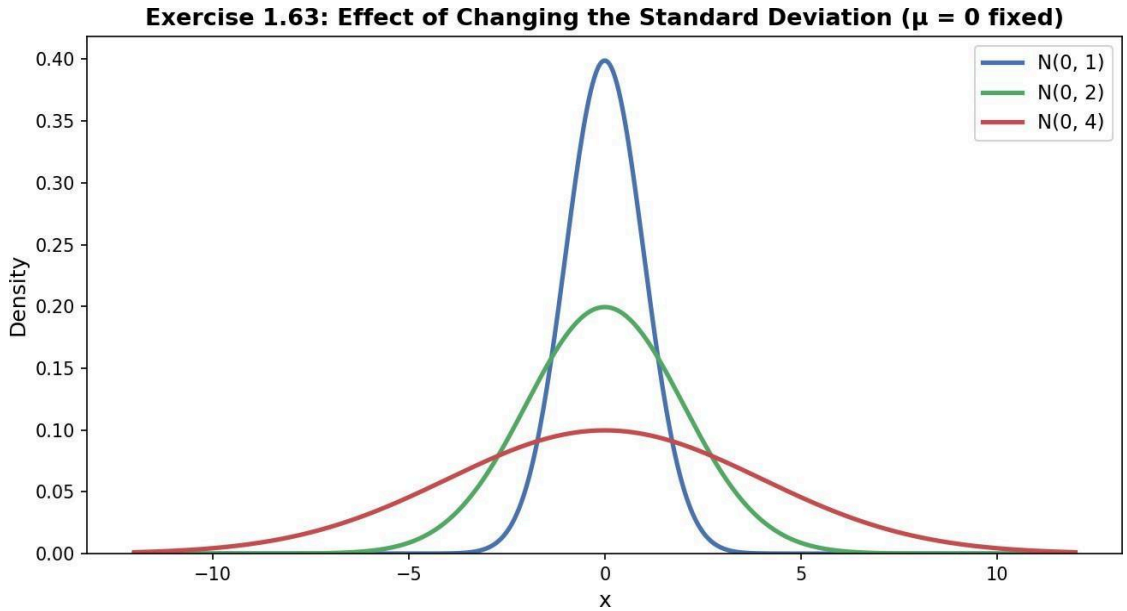
Exercise 1.62: Effect of Changing the Mean

When we increase the mean μ while holding σ fixed, the entire curve shifts to the right along the horizontal axis. The shape and spread of the curve remain identical — only the center moves. Below, three normal curves all have σ = 2, but their means differ (μ = 0, 5, and 10).



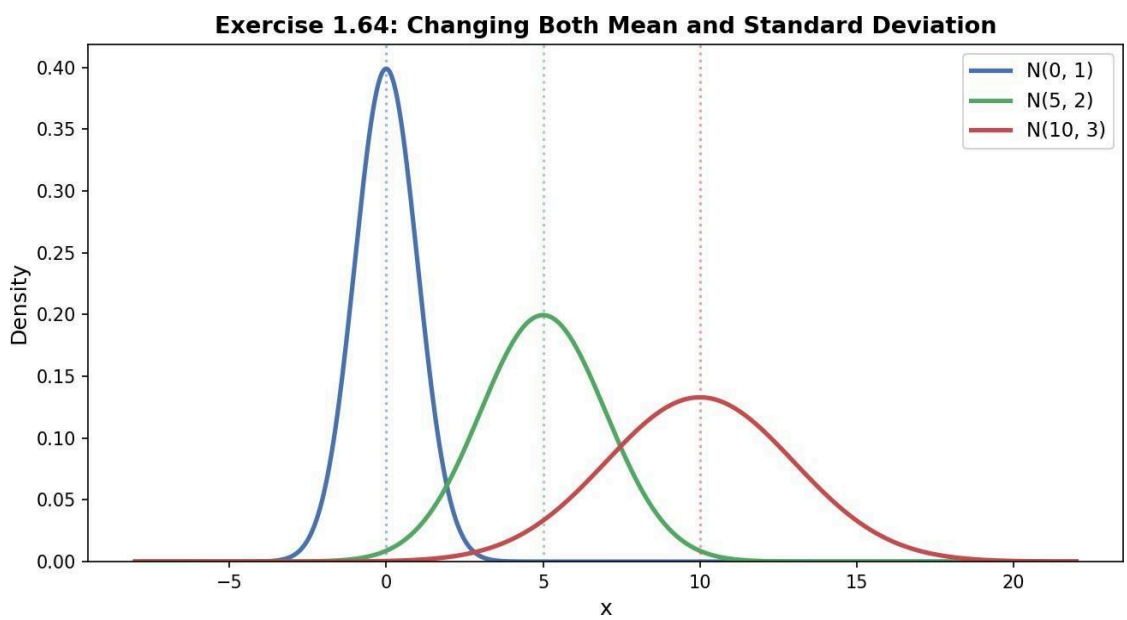
Exercise 1.63: Effect of Changing the Standard Deviation

When we increase the standard deviation σ while holding μ fixed, the curve becomes wider and shorter (flatter). Since the total area under any density curve must equal 1, a wider curve must be shorter. A smaller σ produces a taller, narrower curve concentrated near the mean. Below, three normal curves all have μ = 0, but their standard deviations differ (σ = 1, 2, and 4).



Exercise 1.64: Changing Both Mean and Standard Deviation

When both parameters change simultaneously, the curve both shifts location and changes spread. Increasing μ shifts the curve rightward; increasing σ spreads it out. The combined effect is that the curves differ in both center and shape.



Task 6: Uniform Distributions — IPS Exercises 1.73, 1.74, 1.75

A uniform distribution U(a, b) has a constant density of 1/(b−a) for a ≤ x ≤ b and 0 elsewhere. The probability of any interval is simply the length of that interval divided by the total range.

Exercise 1.73: Uniform(0, 1)

For X ~ U(0, 1): density = 1/(1−0) = 1 for 0 ≤ x ≤ 1.

Mean = (a+b)/2 = (0+1)/2 = 0.5, SD = (b−a)/√12 = 1/√12 ≈ 0.2887

P(0.3 < X < 0.7) = (0.7 − 0.3) × 1 = **0.40**

Exercise 1.74: Uniform(0, 2)

For X ~ U(0, 2): density = 1/(2−0) = 0.5 for 0 ≤ x ≤ 2.

Mean = (0+2)/2 = 1.0, SD = 2/√12 ≈ 0.5774

P(0.5 < X < 1.5) = (1.5 − 0.5) × 0.5 = **0.50**

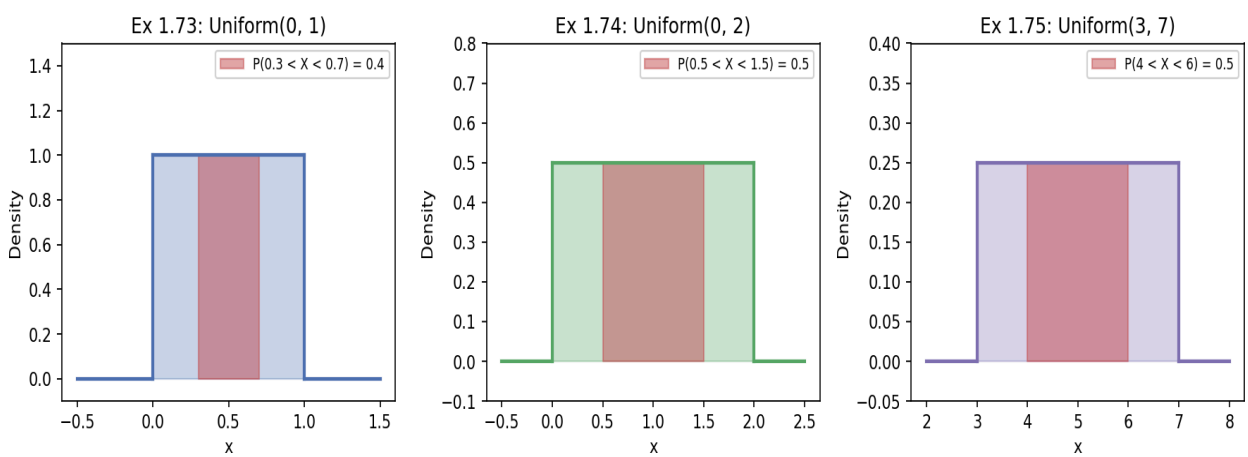
Exercise 1.75: Uniform(3, 7)

For X ~ U(3, 7): density = 1/(7−3) = 0.25 for 3 ≤ x ≤ 7.

Mean = (3+7)/2 = 5.0, SD = 4/√12 ≈ 1.1547

P(4 < X < 6) = (6 − 4) × 0.25 = **0.50**

Task 6: Uniform Distribution Examples



Task 7: Normal Distribution Calculations — IPS Exercises 1.79, 1.80, 1.81

Exercise 1.79: Heights of young women ~ N(64.5, 2.5) inches

(a) $P(X > 68)$: $z = (68 - 64.5)/2.5 = 1.40$. $P(Z > 1.40) = 0.0808$ (8.08%)

(b) $P(X < 60)$: $z = (60 - 64.5)/2.5 = -1.80$. $P(Z < -1.80) = 0.0359$ (3.59%)

(c) $P(62 < X < 67)$: $z_{62} = -1.00$, $z_{67} = 1.00$. $P = 0.6827$ (68.27%)

Exercise 1.80: N(115, 5.6)

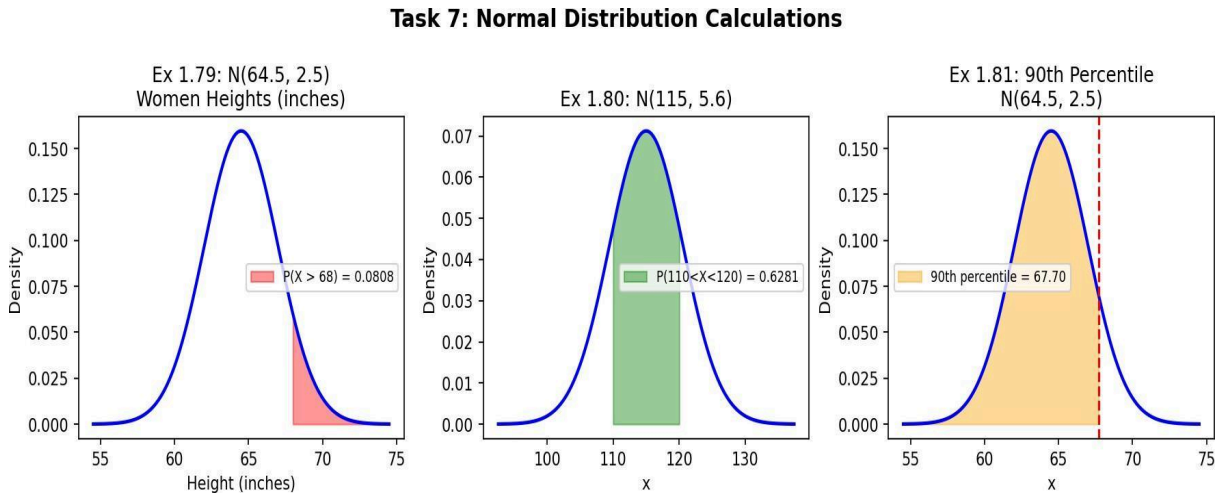
P(110 < X < 120): z_{110} = -0.8929, z_{120} = 0.8929. P = 0.6281 (62.81%)
P(X > 125): z = 1.7857. P = 0.0371 (3.71%)

Exercise 1.81: Finding Percentiles for N(64.5, 2.5)

90th percentile: 67.70 inches (z = 1.2816)

10th percentile: 61.30 inches (z = -1.2816)

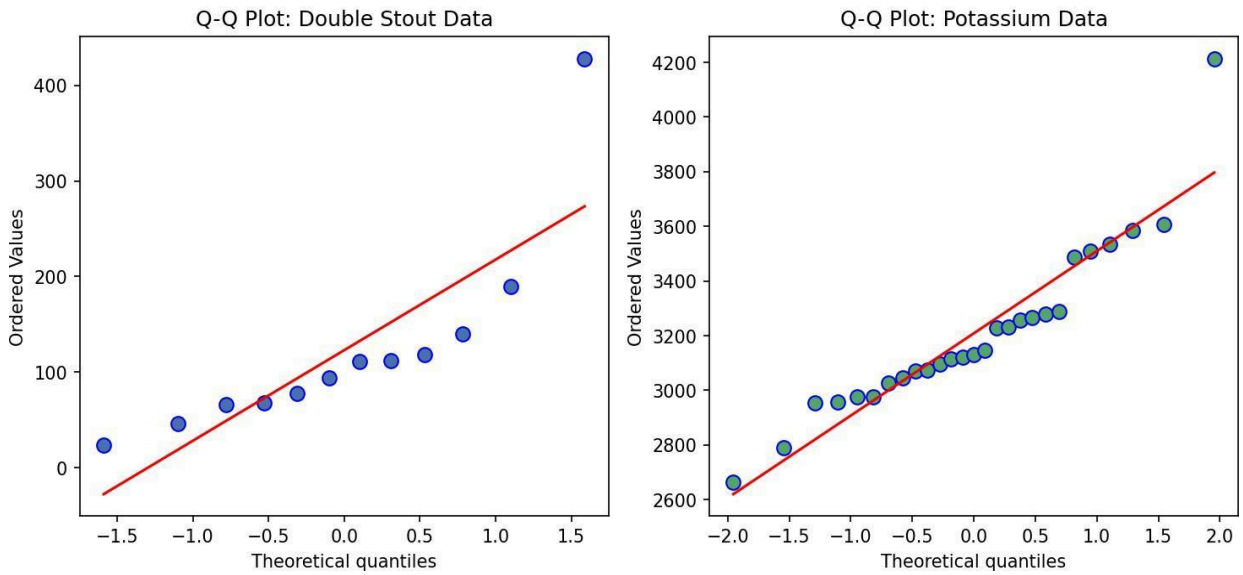
75th percentile: 66.19 inches (z = 0.6745)



Task 8: Normal Quantile Plots

Normal quantile plots (Q-Q plots) compare the observed data quantiles against theoretical quantiles from a normal distribution. If the data are approximately normal, the points will fall close to a straight line.

Task 8: Normal Quantile Plots (Q-Q Plots)



Double Stout Data (left): The Q-Q plot shows a strong departure from normality. The point in the upper right (London = 428) deviates dramatically from the reference line, confirming that this is an extreme outlier. The remaining points show some departure from the line as well, suggesting the distribution is not well-approximated by a normal model even without the outlier. The overall pattern is consistent with a right-skewed distribution.

Potassium Data (right): The Q-Q plot for the potassium data shows points that generally follow the reference line, with some deviation in the upper tail (the observation around 4213 mg). Overall, the data are reasonably consistent with a normal distribution, though the highest value suggests slight right-skewness. The potassium data are much more nearly normal than the double stout data.

Task 9: Do Identical Summary Statistics Mean Identical Distributions?

Question: If two distributions have exactly the same mean and standard deviation, must their histograms have the same shape? If they have the same five-number summary, must their histograms have the same shape?

Answer — Same Mean and Standard Deviation: No, two distributions with the same mean and standard deviation need not have the same shape. For example, a perfectly symmetric bell-shaped distribution and a bimodal distribution with two peaks can share the same mean and standard deviation. The mean locates the center and the standard deviation measures spread, but neither captures the shape (symmetry, number of modes, tail behavior) of the distribution. A classic illustration is Anscombe's quartet, where very different datasets share the same basic summary statistics.

Answer — Same Five-Number Summary: No, two distributions with the same five-number summary (min, Q_1, median, Q_3, max) need not have the same histogram shape. The five-number summary only specifies five values and provides no information about how observations are distributed within each

quarter. For example, two datasets could have the same five-number summary but one could be uniform within each quarter while the other has clusters near the quartile boundaries. The five-number summary captures the overall structure but not the fine detail of the distribution.